Hydro-ecology of the Holocene Green Sahara reveals persistent aridity.

Katie M. Manning[1], Paul S. Breeze[1], Nick A. Drake[1], Adrian Timpson[2]

[1] Department of Geography, King's College London
[2] Research department of Genetics, Evolution and Environment, University College London
Correspondence to: Katie M. Manning (katie.m.manning@kcl.ac.uk)

Abstract

Palaeoenvironmental proxy data have been used to demonstrate regional fluctuations in the climate and palaeohydrology of North Africa during the Holocene. However, the timing and spatial extent of hydroclimatic change and its impact on human and animal populations remains unclear due to considerable incongruity within and between palaeoclimate proxy data and climate models. Here we present a novel comparative assessment of water availability to investigate the extent of greening during the Holocene. We combine palaeohydrological mapping, >1100 radiocarbon dates of palaeohydrological activity, and bioclimatic information from >135,000 faunal remains to demonstrate that the West African Monsoon (WAM) did not extend north of 28°N resulting in a belt of aridity that for much of the Holocene covered a 3° latitudinal band. Even at the maximum extent of both the Mediterranean winter precipitation system and the intensified summer monsoon, a 1° belt of aridity persisted. Furthermore, the WAM was weakened north of 22°N and only extended to 28°N for a short period during the mid-Holocene. This reduced monsoon strength created a mosaic landscape between 22°N and 28°N, that included riparian and desert biomes, rather than an extensive Sahelian landscape that has previously been thought to characterise the Holocene Sahara.

Keywords

African Humid Period; Green Sahara; Biogeography; Palaeohydrology; Climate Change; Palaeoclimate

1. Introduction

The most recent African Humid Period (AHP), otherwise known as the 'Green Sahara', occurred between c. 14,000 and 5000 yrs BP, and was characterised by increased effective precipitation (Tierney et al., 2017), more vegetation (Hély and Lézine, 2014), higher lake levels (Lézine et al., 2011a) and a significant increase in human occupation (Manning and Timpson, 2014). The onset and termination of the AHP is widely recognised to be the result of changes to northern hemisphere insolation resulting from slow shifts in the Earth's orbit which led to an expansion and contraction of the West African Monsoon (WAM) (Demenocal et al., 2000). The transition from a humid climate to widescale aridity at ~5ka is the most recent dramatic example of a climate tipping point during the present interglacial (Hopcroft and Valdes, 2021). However, there remains considerable uncertainty about how the Green Sahara evolved temporally and regionally, and specifically how 'Green' it really was. Gaining clarity on these issues is key to understanding how gradual climate forcing can result in rapid and widespread climate responses and the impact this can have on human and other animal populations both in the past and future.

There are two broad approaches to reconstructing the ecological conditions of northern Africa: proxy records (e.g. marine and terrestrial sedimentary records, and archaeological remains of fauna, flora and pollen) from which environmental changes are directly inferred; and generative climate modelling from which simulations are created. However, there is considerable disagreement both within and between these two approaches (Pausata et al., 2020) such that General Circulation Models (GCMs) fail to reproduce the precipitation increase inferred from palaeoclimate proxy data (Harrison et al., 2015, Perez-Sanz et al., 2014,

Tierney et al., 2017) . Whilst summer insolation increase is known to enhance monsoon circulation and precipitation, a number of additional feedback processes, including dynamic dust (Pausata et al., 2016), vegetation (Lu et al., 2018), and lakes and wetlands (Specht et al., 2022), have been hypothesised to augment GCMs locally, and therefore explain the magnitude of change during the early to mid-Holocene (Claussen et al., 2017). Despite the addition of these feedback processes, however, there is still considerable disagreement in model outputs, which is partly a result of limited validation data, particularly in the northern-most latitudes of the Sahara.

1.1 Spatial extent of the Holocene African Humid Period

Multiple marine core sediment studies from the Mediterranean basin (Ehrmann and Schmiedl, 2021), northwest African coast (Demenocal et al., 2000, McGee et al., 2013) and Gulf of Aden (Tierney and deMenocal, 2013, Tierney et al., 2017) have been used to infer a humid climate as far north as 31°N. However marine records derive their source particulates from poorly defined large regions of the Sahara, which are subject to complex wind patterns before their final deposition. As a result, their interpretation offers widely conflicting views on how far north rainfall penetrated during the AHP. Recent analysis of dust accumulation rates from the Red Sea, suggest that the northeastern Sahara remained arid north of 22°N (Palchan and Torfstein, 2019). In contrast analysis of δD_{wax} records from marine cores along the northwest African coast indicate that monsoon rainfall penetrated as far north as the Atlas Mountains and that the entire Sahara was green, though the correlation between dust flux and δD_{wax} derived precipitation rates is notably weaker in the most northern marine core sediment (e.g. core site GC27) suggesting that the δD_{wax} record might be biased by the Atlas Mountains where precipitation originates from Westerly weather systems (Wassenburg et al., 2016). Other recent work shows that during periods of maximum boreal summer insolation and minimum boreal winter insolation, such as the Early Holocene, the equator-to-pole temperature gradient is weakened in the winter. This drives a southern shift of the Hadley cell, the westerlies, and the Mediterranean storm track, bringing increased winter precipitation to northern Africa (Kutzbach et al., 2014). Using a range of proxy datasets and modelling experiments Wagner and colleagues (Wagner et al., 2019) support this mechanism, proposing that phases of low continental ice volume and high concentrations of atmospheric greenhouse gases, which push North Atlantic low-pressure systems into the Mediterranean and the northern Sahara, can be inferred over the last 1.4 million years. Recent data on the southward extension of the North Atlantic climate system suggests that both the westerlies and monsoons may have contributed to the greening of the desert as far south as the Tibetsi mountains (Hoelzmann et al., 2025). Therefore, whilst these new data, informing on the southward expansion of the North Atlantic climate system, help to reconcile the disparity between Holocene climate model simulations and extant proxy datasets, there remains the question if these two weather systems overlap, or if an arid belt remained between them (Cheddadi et al., 2021).

Unlike marine core sediments, terrestrial proxy records, such as sedimentological and palaeohydrological data, provide important information at specific and well-defined locations. Pollen records demonstrate that savanna vegetation and tropical plants migrated as far as 24°N after which it is argued that the Sahara was composed of a steppe biome (Hély and Lézine, 2014, Jolly et al., 1998). Palaeohydrological evidence meanwhile suggests permanent palaeolakes up to 28°N (Lézine et al., 2011a). These data have also been used to

assess regional heterogeneity in humid tropical conditions. For example, some sub-Saharan aquatic species do not exhibit trans-Saharan spatial distributions, which suggests variation in the extent of surface water within the Sahara (Drake et al., 2011). Patches of relatively arid terrain within a more humid environment, for example, have been hypothesized in parts of the Central Sahara based on pollen core records from Holocene lakes in the region (Hély and Lézine, 2014). A combination of pollen records and stable hydrogen isotope compositions (δD) in leaf wax from lake Tislit in the high Atlas mountains have also been used to infer the source of precipitation (Cheddadi et al., 2021), which the authors suggest is indicative of the Northern Hemisphere's winter storm track and the westerly jet stream.

One way to overcome the discrepancies between paleoclimate proxy datasets is to employ palaeoclimate models, although until recently these have consistently struggled to predict a green Sahara during the most recent AHP, when proxy data are most abundant (Braconnot et al., 2012, Braconnot et al., 2007, Harrison et al., 2015, Liu et al., 2009). Considerable effort has therefore been expended to incorporate new land surface processes into models to provide a more realistic simulation of climate change in semi-arid and arid environments. These studies have found that incorporating lakes or vegetation (Egerer et al., 2016, Specht et al., 2022, Lu et al., 2018) makes the Sahara greener than in previous models, whilst dust and vegetation allows Saharan greening in some models (Pausata et al., 2016, Gaetani et al., 2017), but not in others (Hopcroft and Valdes, 2019). Pausata and colleagues (Pausata et al., 2016), for example, indicate a maximum extent of monsoon rains to 31°N with prescribed vegetation and dynamic dust, compared to 26°N with just prescribed vegetation and only 16°N with orbital forcing alone. Hopcroft and Valdes (Hopcroft and Valdes, 2019), meanwhile, included a fully interactive dust cycle in the HadGEM2-ES model, demonstrating only a small rainfall feedback with dust.

A recent combined analysis of sedimentary data, climate modelling and water balance modelling (Hoelzmann et al., 2025) suggests that strong orographic uplift of moist air masses in the south-central Sahara may have been brought in by north-easterly winds from the Mediterranean, not by the West African monsoon from the south. Whilst the West African monsoon appears to have reached the Tibesti mountain region during the early Holocene, Hoelzmann and colleagues (Hoelzmann et al., 2025) argue that it was too weak to be uplifted to the higher crater lakes, located around 21°N and 16.5°E. However, the underpinning climate model in this analysis fails to capture observed wetness in more northern regions of the Sahara e.g. Takakori in the Tadrart Acacus at 24.5°N (Cremaschi et al., 2014), where wet environmental conditions are evidenced by pollen, fish remains and geomorphological features, suggesting a source of error in the simulated moisture fluxes from Jungandreas et al. (Jungandreas et al., 2023, Jungandreas et al., 2021). Similarly, Cheddadi and colleagues present a set of idealised scenarios which incorporate a hypothesised extension of the Westerlies to at least 24° N and potentially 18° N based on the lake core evidence from Lake Tislit. They achieve a 'green' Sahara throughout the Holocene, which they argue shows that the Westerlies and WAM 'must have met and possibly overlapped' (Cheddadi et al., 2021). However, this conceptual framework lacks a formal validation test and therefore relies on the assumption that the entire Sahara was green over the course of the Holocene. Therefore, whilst new proxy data provides valuable insight into the southward expansion of the North Atlantic climate system and the idealised model parameters help to reconcile the disparity between Holocene climate model simulations and the extant proxy datasets, the questions remain regarding if and where these two weather systems overlap.

1.2 Timing and abruptness of the Holocene African Humid Period

Marine core sediments from the northwest African coast have also been used to quantify the onset and termination of humid conditions, indicating a rapid hydroclimatic shift between 12,500 -11,500 and 5500 years BP. Whilst it is known that these records represent 'Saharan' climate change (e.g. (Demenocal et al., 2000, Tjallingii et al., 2008) the location of the desert dust sources found in marine records are still debated. The Bodélé Depression located at 17°N, for example, is believed to emit 0.7 million tons of dust per day (Koren et al., 2006), making it the world's greatest single dust source. It also shows a rapid drying event at 5000 BP (Armitage et al., 2015), which aligns well with the marine record from ODP core 658C (Adkins et al., 2006, Demenocal et al., 2000). In contrast, pollen and sedimentological records from Lake Yoa in northern Chad suggest that the termination of humid conditions may have been gradual and progressive (Francus et al., 2013, Kröpelin et al., 2008) although the contribution of groundwater to the ecological conditions surrounding Lake Yoa complicate the situation. Although these lakes are remnants of a much larger lake, they are still in existence today and are almost exclusively fed by permanent groundwater outflow from a regional aquifer. Similarly, some of the northern palaeolakes e.g. Hassi el Mejnah and Sebkha Mellala display larger water volumes during the AHP (Gasse et al., 1990, Gasse and Van Campo, 1994), yet they probably reflect a component of groundwater discharge which is recharged at the M'zab mountains, rather than direct precipitation (Gasse et al., 1990). Sebkha Mellala, for example, still receives runoff from the Atlas Mountains to the north of the Sahara today.

This paper addresses the core question of how far north the WAM extended, and the extent to which the Westerlies contributed to the Holocene greening of the Sahara. We use updated trans-Saharan palaeohydrological data alongside a novel bioclimatic indicator, using faunal records from archaeological sites to produce a comparative assessment of water availability through time and space. Faunal remains provide valuable biogeographic information to help understand changing patterns of vicariance, climate and habitat, that when combined with the palaeohydrological data creates a more detailed and robust understanding of the past Saharan hydroclimate.

2. Materials

We performed five distinct analyses that utilised three core datasets - remote sensing data, and 14C dates from both palaeohydrological samples and faunal records (Fig. 1 and Fig. 2). All data was collected as part of the Leverhulme-funded Green Sahara project, which produced a spatial database of Holocene archaeological and palaeoenvironmental data, including occupation details, 14C data, palaeohydrology, pollen, faunal remains, carbon isotopes and rock art. Because modern precipitation records indicate that at the decadal scale, both the WAM and Westerlies produce a remarkably delineated boundary between arid and wet across north Africa, (Fig S1), with the WAM generating a clear horizontal delineation, we can assume that at the broadest scale, there was similar wetness at any latitude, during the Holocene. This assumption allows multiple proxy records to be aggregated latitudinally to leverage sample size when estimating latitudinal variation in wetness. Furthermore, this assumption simplifies the central research question of what the latitudinal extent of the green Sahara was.

To assess the spatio-temporal trends we mapped the palaeohydrological features across the region and used published palaeoenvironmental data to determine the evidence for regional wetness between 15ka and 2ka. For the palaeohydrological mapping we analysed all dates within catchments which extend from beyond the 200mm Isohyet into the Sahara, to capture where endogenous moisture transport could have altered the desert conditions (the Nile and Niger being the most obvious examples). For all other analysis we defined our study area of interest as everything within the 200mm Isohyet, typically considered to be the 'aridity line' characterising the modern-day Sahara. All samples from the Nile Valley were excluded to avoid conflating permanent hydrological activity along the river with monsoon-fed hydrological re-activation. We therefore used a total of 1323 C14 dates from 515 hydrological sites, 3767 C14 dates from 1660 archaeological sites, and 135,917 faunal remains from 142 sites phases, from 118 archaeological sites (Fig. 2) dated between 15ka and 2ka. Details of the methods can be found below in the respective analytical section.

3. Spatial and temporal trends in the hydro-ecology

3.1 Palaeohydrological mapping

Palaeohydrology was mapped using remote sensing and GIS (Geographic Information System) techniques, applying the methodology developed by Breeze et al. (Breeze et al., 2015) to identify relict drainage systems, lakes and wetlands and the method of Drake et al. (Drake et al., 2011) to map the fluvial fans across more than 9 million km^2 of northern Africa. Flow accumulation data were calculated from HydroSHEDS 3AS data (Lehner et al., 2008) to map major surface palaeodrainage at ~90 m resolution, with river networks being classified by thresholding, with contributing upstream flow accumulation areas in excess of 1000 km^2. Associated river catchments were calculated using HydroBASINS (Lehner and Grill, 2013) This mapping shows that the Sahara is covered with closed basins, containing paleolakes, a dense network of palaeochannels, and large fluvial fans with extensive distributary channel networks (Fig. 1).

The identified palaeohydrological features represent aggregated signatures from multiple greening events which created and re-activated local hydrological systems during the course of the Quaternary. Thus, in order to identify which areas where likely to have been activated during the most recent AHP, we combine radiocarbon dates from samples directly associated with hydrological activity, with catchment reconstructions that use HydroBASINS (Lehner and Grill, 2013), which delineates nested watersheds based on Pfaf-stetter ordering. After identifying which catchment contains these hydrological C^{14} dates, we generate an overlay map of activated catchments in discrete time intervals, providing a spatiotemporal wetness proxy for the entire Holocene (Fig. 2). This approach suggests some amount of catchment activation occurred across almost the entire Sahara between 15 and 2ka, with only small areas in central Algeria, Northern Mauritania and Eastern Libya showing no evidence of hydrological activity. The Tibesti and neighbouring catchments show an interesting pattern of repeated activation throughout the Holocene (11/12 out of the 13 time slices), supporting the recent analysis by Hoelzman and colleagues (Hoelzmann et al., 2025). The distribution of re-activated catchments also aligns well with the earlier identification by Drake et al (Drake et al., 2011) of key megalake

basins and large alluvial fans, namely Lake Megachad and the adjacent Sahabi River, as well as the Niger Basin and the West Ahaggar Mountains fluvial fan (numbers 1 - 4 on Fig. 1, respectively).

Whilst this catchment analysis provides a strong proxy for temporal variation in wetness, the spatial component of this proxy has a number of limitations. Firstly, rainfall in any part of a catchment will flow to the same accumulation point (such as a river or lake), allowing the direct inference that water at the accumulation point came from somewhere upstream in the catchment. However, in deserts, catchments are typically characterised by endorheic basins hosting one or more terminal lakes rather than the throughput to ocean outlets more typically seen in wetter climes. In such cases multiple drainage systems terminate in the same depression, and it is possible for rainfall to only occur in one portion of a catchment and still generate a lake at its core. Likewise, although a dated river demonstrates upstream flow, precisely where upstream that water originated cannot be determined. Thus, identifying precisely where in a catchment the dated rainfall came from is impossible without spatially extensive on-the ground sedimentological provenancing, which is currently lacking for the Sahara. For this reason, we can only use the catchment itself as a unit of interpolation. A further challenge is that while having discrete divides at any given scale, catchments are fractal, consisting of numerous nested sub-catchments, and thus can be calculated at a variety of scales, despite them often in reality being part of a more diffuse hydrological system (Brooks et al., 2015). We attempted to mitigate the above factors and the possibility of confounding effects if multiple high mountain areas (and hence orographic moisture sources) were included in the bounds of a single catchment. We utilised the Hydrobasins data to subdivide the largest level 4 endorheic catchments into sub-units with shared mountain sources. This helped mitigate against a scenario where fluvial sources on one side of an endorheic basin, fed from one set of mountains, would artificially indicate a presence of moisture on the opposite side where drainage came from other mountains. However, the accumulator problem of large central lakes still applies. Nevertheless, when we map the evidence for human populations occupying the Holocene Green Sahara, there is good correspondence between the density of occupations sites and activated catchments (Fig. 3). In particular, there is good correspondence between the evidence for human occupation and the re-activated catchments that connect the Niger basin with the Ahnet-Mouydir Basin/Chotts basin via the Irharhar River. This re-activation of hydrological catchments occurred between 3 and 5 times (within 1000 year intervals) during the early Holocene, suggesting a corridor across the Sahara that likely occurred when the WAM was more intense(Marzin and Braconnot, 2009, Rachmayani et al., 2015, Kutzbach et al., 2020, Duque-Villegas et al., 2024). The evidence from these archaeological sites, however, are very sparse and likely represent ephemeral campsites resulting from high mobility, rather than permanent occupation sites.

Whilst palaeohydrological reconstruction can provide information about the spatial extent of hydrological reactivation, they do not necessarily shed light on regional ecological conditions, nor do they provide information about the source of precipitation in respect to different circulation systems. Both large exogenous river systems, such as the Niger and Nile. and very large lakes, such as Lake Chad, are well known to traverse desert environments, originating outside the desert and creating riparian environments that provide corridors through what is otherwise an arid environment. Though these environments are often spatially limited and thus do not support large populations of savanna species, they can allow for dispersal of some more water-dependent species deep into the desert. To investigate the broader regional ecological conditions across the

whole Sahara, we use faunal remains from archaeological sites. These provide observational evidence of local animal populations at a given point in time and space and can be used to infer a regional wetness index that can be compared directly with the palaeohydrological data.

3.2 Faunal remains as a bioclimatic indicator

Because the composition of local-scale animal communities depends upon the regional species pool, and ecological constraints of the environment, most notably vegetation and surface water availability, we can use this data as a bioclimatic indicator. We develop a Water Dependency Index (WDI), which is based on three criteria – maximum number of days a species can survive without access to water, the dependency on permanent water, and how geographically constrained the species is, only using species that are widely distributed. As such, the WDI can be used to infer whether there was sufficient surface water to support a regional population of water-dependent species, or conversely if the species do not require water and aridity exists. Whilst 628 different animal taxa were identified, only 85 of the taxa have distinct ecological signatures (e.g. they are not broad-spectrum species and have specific water requirements). Table 1 shows the representation of taxa and their relative WDI.

WDI	**Hydro-ecological indicator**	**Taxa (n=number of species included)**
1	Permanent water/0 days. Extreme geographic constraint	Fish (n=59); Crocodile (*Crocodylus niloticus/Corocodylus sp.*), Hippo (*Hippopotamus amphibius*)
2	Water dependent/1 day; High geographic constraint	Elephant (Loxodonta africana); Waterbuck (Kobus ellipsiprymnus); Kob (Kobus kob); Black Wildebeast (Connochaetes gnou); common patas monkey (Erythrocebus patas)
3	Savanna with dispersed permanent water; Moderate geographic constraint	Lion (*Panthera leo*), Rhino (*Rhinocerotidae/Diceros or Cerathotherium*); Black rhino (Diceros bicornis); White rhino (*Ceratotherium simum*); Giraffe (*Giraffa camelopardalis*); Topi (*Damaliscus lunatus*);
4	Dry savanna/ Seasonal water; Low geographic constraint	Ostrich (*Struthio camelus*); African wild ass (*Equus africanus*); Dama Gazelle (*Gazella dama*); Sand cat (*Felis margarita*); Hartebeest (*Alcelaphus buselaphus*); Thomson's gazelle (*Eudorcas thomsonii*)
5	Desert/stony plains /Little to no water; No geographic constraint	Cape hare (*Lepus capensis*); Dorcas gazelle (*Gazella dorcas*); Barbary sheep (*Ammotragus lervia*); Oryx (*Oryx dama*)

Table 1. Hydro-ecologically distinctive taxa used in the WDI analysis.

Throughout the Holocene, water-dependent animals requiring local permanent water, such as hippo (*Hippopotamus amphibious*; WDI=1), and crocodile (*Crocodylus niloticus*/sp.; WDI=1) are absent beyond 21°N (Fig. 3). Other species that are dependent on more dispersed water sources, requiring water every 1-4 days, such

as elephant (WDI=2) and giraffe (WDI=3), appear to extend to 24.2°N (note the elephant at 33°N is from a single bone at Haua Fteah on the Mediterranean coast and dates to the Late Pleistocene). In contrast, dry Savanna, xerocole species such as the hartebeest (*Alcelaphus buselpahus*; WDI=4), ostrich (*Struthio camelus*; WDI=4), and Barbary sheep (*Ammotragus lervia*; WDI=5), become increasingly common north of 25°N, and are prevalent at sites located between 29°N and 37°N (Fig. 3). The hartebeest is an adaptive dry savanna species, which has a particular preference for medium grasslands dominated by red oat grass (*Themeda triandra*) and scrub acacia (Estes, 2012). Whilst the hartebeest will drink regularly when freshwater is available, it is also capable of eating melons and digging up roots to meet its needs. This is observed in hartebeest that occupy the Kalahari today (Knight, 1995). Ostrich, which typically occupy Savanna and desert environments, have evolved digestive systems which allow them to source more water and nutrients from consumed forage (Skadhauge et al., 1984). Similarly, the barbary sheep are found in arid environments, particularly mountainous areas, where they graze and browse grasses, bushes, and lichens and can satisfy their metabolic water requirements from food alone, although like the hartebeest, they will take advantage of surface water if it is available (Mimoun et al., 2017). The presence of xerocole species alongside more water-dependent species in southern latitudes, is therefore not indicative of aridity, and probably reflects a shift in modern-day species distribution due to anthropogenic disturbance. However, in the absence of water-dependent species, the presence of only xerocole animals suggests there was limited surface water to support more water-dependent communities, indicating the predominance of broad scale dry savannas and arid mountains north of 25°N. Previous analysis of the zoogeography of North Africa (Drake et al., 2011) suggested that savanna species such as elephant (*Loxodonta africana*) and giraffe (*Giraffa camelopardalis*) could have extended further north up to 29°N. However, these records are based on rock art depictions, which introduces a number of potential biases, including uncertainty on dating, and whether the cultural perceptions of the Holocene artists got their inspiration from the regional macrofauna, or whether they serve the formation and recursive maintenance of ancestral memory. Such remote mapping of cultural geography has been observed ethnographically across many desert environments (Weiss, 2005, McDonald and Veth, 2013, Porr and Bell, 2012).

Figure 4 demonstrates a clear shift from mixed communities, including both fully water-dependent and semi-arid/arid between 15°N and 27°N, to communities made up entirely of semi-arid and arid species north of 27°N. Importantly this agrees with the archaeological date where there is a gap between 31°N and 33°N, where no archaeological sites (excluding coastal sites that are above the 200mm isohyet) have yielded faunal remains. Several surface deposits, mostly composed of lithics and dating to the epipalaeolithic, have been identified around Ouargla in Algeria and along the Jebel Nafusa in Libya, but these yielded no faunal remains. Whilst this absence of evidence is not evidence of Holocene populations being absent from these latitudes, there is no substantive reason for why animal communities would not be represented in archaeological assemblages, unless occupation was so ephemeral as to negate the deposition of faunal remains. Deflation of archaeological deposits and desiccation of organic remains are a problem for preservation across the Sahara, and not just in the most northern latitudes. Equally, soil acidity shows no latitudinal structure (Jones et al., 2013) that would explain the differential absence of faunal remains between 31°N and 33°N.

Using the full archaeozoological data from our study area, comprising 142 site phases, we can also evaluate changes in species diversity across space and time, allowing us to evaluate the relationship between biodiversity and hydrology. Because hydrological processes drive biodiversity by influencing factors such as

water quantity and quality, nutrient availability, and niche construction, a measure on species diversity can shed light on the carrying capacity and overall ecosystem health of the region. We use two biodiversity metrics – Shannon and Richness, shown in Fig. 4, which reveal a clear latitudinal decline in species diversity. The decline in diversity corresponds well with the WDI values – at lower latitudes site phases have a low WDI value and there is a wide range of species, whilst site phases in more northern latitudes have a higher WDI value and lower species diversity.

3.3 Hydro-ecological correspondence

In addition to using the hydrological C^{14} dates in the catchment analysis, we can also assess the latitudinal extent of hydrological date density using the mean calibrated date plotted on a kernel density surface (Fig. 5). This allows direct comparison to the WDI and reveals a high density of radiocarbon dates associated with hydrological activation in the early Holocene (11ka to 7ka) up to 22°N, which correlates well with the distribution of sites with low WDI values. Around 7 ka, there is evidence for a transition toward drier climatic conditions, marked by a pronounced decline in hydrological activity. Nevertheless, mean WDI values indicate that adequate surface water persisted as far south as 22° N throughout the Holocene. Unsurprisingly, the spatio-temporal reconstruction of surface wetness shown in Fig. 5 corresponds closely with what has already been presented by Lezine et al. (2011) due to being based largely on the same dataset. However, the connection to the animal WDI values provides additional insight. Just north of 22°N, there is a clear predominance of arid-adapted species. The presence of more water-dependent species around 25°N come from the Tadrart Acacus Mountains of southwestern Libya, namely the sites of Takakori and T-n-Torah, where both fish assemblages and crocodile were present throughout the Holocene. The single WDI value of 3 located at 26°N comes from Edeyen Murzuq 94/18, where the only two taxa identified to species level were the bohor reedbuck (*redunca redunca*) and the hartebeest (*alcelaphus buselpahus*). Other, more arid adapted species include indeterminate gazelle and oryx, and were noted as present, although no quantitative analysis of the faunal remains was undertaken. Otherwise, North of the Tadrat Acacus, the faunal assemblages include only arid-adapted animal communities and are notably lacking in any sub-Saharan species.

Critically, Fig. 5, reveals a clear band of aridity north of 28°N. For much of the Holocene, this arid belt spreads across almost 3° of latitude between 27°N and 31°N and is only reduced to a narrower 1° latitudinal width for a short period of time, between c. 7.2ka and 5.5ka. Hence, we propose that this demarcation in the evidence for wetness, in both the palaeohydrological dates and WDI values, represents the southern-most extent of the Westerlies and the northern reach of the WAM. Beyond 22°N, however, the density of surface water is clearly reduced and we can observe a clear delineation in both the hydrological date density and the WDI values, which at this latitude averages 4.4, indicative of a persistent dry savanna. This abrupt latitudinal boundary indicates a rapid decline in the WAM response north of 22°N. Above this, smaller regional ecologies which supported more water-dependent species occur only in sporadic areas as a consequence of complex interactions of groundwater recharge, and orographic uplift in mountain regions. The absence of sub-Saharan species north of 25.14°N (e.g. Tilapinii and Claridae) further demonstrates the WAM response being significantly weakened in these northern regions as to negate dispersal of sub-Saharan species beyond this limit.

4. Discussion and conclusion

Deterministic models of Holocene greening across North Africa have significantly improved over recent years, incorporating dynamic dust (Pausata et al., 2016, Hopcroft and Valdes, 2019), vegetation feedbacks (Lu et al., 2018), and moisture recycling from lakes and wetlands (Specht et al., 2022). Whilst 'orbital-only' models typically stalled the WAM extent at ~17–20°N, most coupled climate simulations extend the monsoon rain belt northward to ~26°N. Previous attempts at model validation have typically relied on course temporal and spatial datasets, notably vegetation reconstructions from pollen (Hély and Lézine, 2014, Kröpelin et al., 2008, Lézine et al., 2011b) and leaf wax biomarkers (Castañeda et al., 2009, Tierney et al., 2017). Yet, these proxy datasets fail to provide a comprehensive spatio-temporal assessment of regional wetness. Depending on whether exclusive, or non-exclusive taxa are used, pollen records suggest the Sudanian group, which require rainfall rates between 500 and 1500 mm/year, expanded either to just 14-16°N or 25°N respectively during the early Holocene (using the 0.85 isoprobability delineating the maximum extent zone in which 85% of the samples are included (Lézine et al., 2011a) In contrast, based on leaf wax biomarkers and inferred precipitation rates from ocean core sediment sites, Tierney et al. (Tierney et al., 2017) propose that a seasonal tropical climate dominated most regions of North Africa during the Holocene. Our combined hydro-ecological datasets provide a much higher spatial and temporal resolution than previously available and help to shed light not only on the northern extension and strength of the WAM, but also on the southern reach of the Westerlies.

Using an updated radiocarbon dataset from palaeohydrological sites, combined with a new wetness indicator based on the water-dependence of the animal populations that were hunted by Holocene occupants of the Sahara, we propose that the WAM did not extend beyond 28°N and was significantly weakened north of 22°N. Neither the animal data nor palaeohydrology provides evidence of the westerlies extending south of 29°N resulting in a belt of aridity that for much of the Holocene covered a 3° latitudinal band. Even at the maximum extent of both the Mediterranean winter precipitation system and the intensified summer monsoon, a 1° belt of aridity persisted across all latitudes. Instead of a seasonal tropical climate, our data lends support to previous pollen and palaeohydrological reconstructions that indicate a mosaic landscape of dry Savanna and more humid mountainous regions, with the latter receiving sporadic changes in precipitation throughout the Holocene.

Gaining clarity on the spatial and temporal extent of these Holocene humid conditions has major implications for understanding both regional environmental change and global climate feedbacks. The Sahara is a critical component of the Earth system, influencing atmospheric circulation, dust emission, and oceanic biogeochemistry. Reconstructing its hydrological and ecological evolution provides a key analogue for future climate trajectories, showing how gradual orbital forcing can generate abrupt, regionally variable responses. The persistence of aridity north of 27°N, even under peak insolation forcing, underscores the non-linear nature of monsoon–desert interactions and the existence of strong internal thresholds within the North African climate system.

Future work should expand this terrestrial proxy coverage, especially in the central and northern Sahara, and integrate it with palaeoceanographic, isotopic, and speleothem records from adjacent regions. Such multi-proxy syntheses will refine reconstructions of the latitudinal hydrological gradient between the tropics and

the Mediterranean and help constrain the phasing and strength of the WAM and Westerlies through time. Coupling these datasets with next-generation Earth system models will also improve understanding of teleconnections between the Saharan hydroclimate and other components of the earth system, such as intertropical convergence zone migration, and carbon–dust cycle feedbacks. Given the Sahara's disproportionate role in modulating global albedo, atmospheric dust, and tropical–extratropical circulation, resolving the limits and drivers of its Holocene greening is essential for constraining the stability and sensitivity of the Earth's climate system under future warming scenarios.

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Data availability

All data and code is available at AdrianTimpson/greenSahara: Analysis code and data to accompany submission

Author Contributions

Katie Manning: Conceptualization, Methodology, Visualization, Data curation, Writing – original draft, Writing- Reviewing and Editing. **Paul Breeze**: Methodology, Visualization, **Nick Drake**: Writing- Reviewing and Editing. **Adrian Timpson**: Methodology, Visualization, Writing- Reviewing and Editing.

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